

FUP - Oral Exam

neslusam and question providers

Contents

Introduction	1
Functional Programming	1
Racket	1
λ -Calculus	1
Haskell	2

⚠ TRIGGER WARNING - M*NAD ⚠ - The text below contains repeated, explicit occurrences of the “M-Word”. If you are currently recovering from trying to understand what a m*nad actually is, further review of this material is not recommended..

Introduction

[V] - Tomáš Votroubek , [H] - Rostislav Horčík , [Z] - Matěj Zorek **Your examiner is random, your previous lecturer doesn’t make a difference.**

There might be a newer version of this document, check out this [github repository](#)

Functional Programming

[Z] *What is pure/Higher-order function + examples of each possible combination?*

[Z] *List types of recursions based on branching factor and provide examples. Classify head/tail recursion and discuss theirs branching factor.*

Racket

[V] *How to implement infinite stream of nats/ones*

- 1. using stream functions?*
- 2. without using stream functions?*

[V] *Describe how does foldl/foldr work, which parametres are used and what is the difference between foldl and foldr.*

λ -Calculus

[Z] *List and describe terms in syntax of λ -calculus.*

[H/Z] What is a redex?

[H] What is a reduction? List and describe different types of reductions and their usage.

[V] What is a normal form and how to transform a λ -expression to its normal form?

[Z] What is the difference between bound and free variable? Which operation is done on them?

[H] What are the Church-Rosser theorems?

[H] how to represent True/False in λ -Calculus?

[H] How to represent number in λ -Calculus? How to represent addition of 2 numbers?

[H] reduce this expression $(\lambda x.x(\lambda x.xx))e$ to its normal form.

[V] Find all free/bound vars , redexes in $(\lambda f. (\lambda x. f (x x)) (\lambda x. f (x x)))$. Find normal form.

Haskell

[V] Describe signature of function map.

[V] Describe signature of function foldl/r.

[H] What is a typeclass?

[H] How to implement polymorphic typeclass , which adds 1 to an Int and prints out length for String?

[V] What is Functor typeclass? Which functions define it?

[V] What is a Applicative typeclass? Which functions define it?

[V] What is a Monad? Which functions define it?

[H] What is a State Monad, what is it good for? How to implement bind for State Monad?

[H] What are semigroups, monoids¹ and foldables?

You are all done!



¹**Monoids** are not to be confused with **monads**. That said, a monad is just a monoid in the category of endofunctors but one need not understand that in order to understand monoids and monads from a programming point of view.